US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250283550
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	YOSHIDA; Tatsuya et al.

ELECTRICALLY OPERATED VALVE

Abstract

In an electrically operated valve including a valve main body having a valve chamber communicating with an inflow passage for refrigerant and an outflow passage for refrigerant, a valve body moving forward and backward for a valve seat formed within the valve chamber and changing a refrigerant flow rate, an electric motor driving the valve body, a board controlling the electric motor, and a board storage portion including a case storing the board, the board has a fixing hole fixing the board within the case, the case has a support projection fixed into the fixing hole and supporting the board within the case, and the support projection is integrally formed with the case portion. The support projection is preferably provided with a step portion and the board is preferably held by the step portion and a pressing projection disposed in a back surface of a lid occluding the case.

Inventors: **YOSHIDA; Tatsuya (Tokyo, JP), HIRATSUKA; Masaki (Tokyo, JP)**

Applicant: **FUJIKOKI CORPORATION** (Tokyo, JP)

Family ID: **86335501**

Appl. No.: **19/216885**

Filed: **May 23, 2025**

Foreign Application Priority Data

JP	2021-183278	Nov. 10, 2021

Related U.S. Application Data

parent US continuation 18250820 20230427 parent-grant-document US 12331842 WO continuation PCT/JP2022/033522 20220907 child US 19216885

Publication Classification

Int. Cl.: **F16K31/04** (20060101); **B60T8/36** (20060101); **F16K27/02** (20060101)

U.S. Cl.:

CPC **F16K31/046** (20130101); **F16K27/029** (20130101); B60T8/3675 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/250,820 filed on Apr. 27, 2023, which is a 371 of PCT/JP2022/033522 filed on Sep. 7, 2022, which, in turn, claimed the priority of Japanese Patent Application No. 2021-183278 filed on Nov. 10, 2021, and the above applications are incorporated herein by reference.

TECHNICAL FIELD
[0002] The present invention relates to an electrically operated valve, and more particularly to an electrically operated valve including a board which controls an electric motor, and a resin case which stores the board.
BACKGROUND ART
[0003] An electrically operated valve controlling an opening degree of a valve with the use of an electric motor such as a stepping motor has been conventionally used for a refrigeration cycle apparatus which is provided with a refrigerant circuit such as an air conditioning machine and a refrigeration/freezing apparatus.
[0004] Further, as the electrically operated valve as described above, there is an electrically operated valve including a controller which controls an exciting current of a coil. The controller is mounted to a printed board, and the board (refer to “control board” or simply to “board” in the present application) is stored in a resin case which is disposed near a stator.
[0005] Further, the following patent literature 1 exists as a publication disclosing the electrically operated valve as described above.
CITATION LIST
Patent Literature
[0006] Patent Literature 1: Japanese Unexamined Patent Publication No. 2021-110409
SUMMARY OF INVENTION
[0007] In the meantime, in the electrically operated valve provided with the control board as mentioned above, the board has been attached by being screwed or caulked to a support portion which is disposed in the resin case, and there is left room for improvement about a structure for supporting the board. The attaching work by screwing or caulking can not be always said to have a good work efficiency. On the contrary, if the structure for supporting the board can be simplified, reduction of a parts number and an assembling step number can be caused. Therefore, it is possible to reduce a manufacturing cost of the electrically operated valve.
[0008] Further, in this kind of electrically operated valve, in order to prevent infiltration of moisture, the electric motor (a stator including a coil and a yoke) driving a valve body is covered with a resin cover (refer to “outer shell forming portion” or “outer shell cover” in the present invention). However, the resin case storing the board has been conventionally formed separately from the outer shell cover, and has been fixed to the outer shell cover (refer to the above-described patent literature 1). Therefore, with regard to these resin forming portion (outer shell cover and resin case), there is room for reduction in the manufacturing step number.
[0009] Accordingly, an object of the present invention is to allow a board to be attached to an electrically operated valve with good workability, and to reduce a parts number and a

manufacturing step number of the electrically operated valve with board.

[0010] In order to solve the problem mentioned above and achieve the object, an electrically operated valve according to the present invention is an electrically operated valve including a valve main body having a valve chamber which is communicated with an inflow passage introducing a refrigerant and an outflow passage discharging the refrigerant, a valve body which changes a flow rate of the refrigerant by moving forward and backward with respect to a valve seat formed within the valve chamber between a valve close state seating on the valve seat and a valve open state separating from the valve seat, an electric motor which drives the valve body, a control board on which an electronic component controlling the electric motor is mounted, and a board storage portion which includes a case portion storing the control board, wherein the control board has a fixing hole which fixes the control board to an inner side of the case portion, the case portion has in an internal portion thereof a support projection which supports the control board within the case portion by being inserted into the fixing hole, and the support projection is integrally formed with the case portion.

[0011] The electrically operated valve according to the present invention is structured such that the control board is supported to the inner side of the case portion by inserting the support projection disposed within the case portion into the fixing hole of the control board without screwing or caulking. Therefore, according to the present invention, the board can be installed within the case only by slotting the support projection to the case portion while inserting the support projection into the fixing hole, the screw for supporting the board is not required, and the screwing work or the caulking work is not required. Thus, it is possible to reduce the parts number and the manufacturing step number. Further, a time for forming and attaching the support portion for supporting the board separately from the case portion is not required since the support portion is integrally formed with the case portion.

[0012] Further, the electrically operated valve according to the present invention may be provided with an outer shell forming portion which covers a stator of the electric motor, and the case portion may be integrally formed with the outer shell forming portion. According to the aspect mentioned above, it is possible to omit a work for manufacturing the case portion separately from the outer shell forming portion and fixing.

[0013] Further, in the electrically operated valve mentioned above, a coil coating forming portion covering a coil provided within the stator may be disposed in an inner side of the outer shell forming portion. According to the aspect as mentioned above, by performing two stages of forming steps (that is, constructing as a double coating structure), it is possible to prevent a winding wire of the coil from being damaged when forming, and it is possible to avoid void generation in the outer shell forming portion having a complex shape and reduction of strength. More specifically, it is possible to simultaneously achieve the winding wire breakage prevention when forming and the strength securement of the forming portion by forming at a low pressure the coil coating forming portion covering the coil which tends to be generated the breakage such as deformation, insulation failure and breaking of wire due to resin pressure application when forming, and thereafter forming at a high pressure the outer shell forming portion covering the coil coating forming portion.

[0014] In the meantime, the outer shell forming portion is preferably formed by introducing the resin from a gate (introducing port) at one position in such a manner as to prevent a weld line causing the reduction of strength from being generated. This is because of preventing a crack from being generated in the weld line due to the temperature change when aging use and preventing the moisture from infiltrating. On the contrary, when employing the double coating structure (structure provided with the coil coating forming portion in the inner side of the outer shell forming portion) as mentioned above, it is not particularly necessary to take into consideration the strength in the coil coating forming portion in which the strength can be secured by being covered by the outer shell forming portion. Therefore, it is possible to more securely cover the coil having the complex shape by introducing the resin from the gates at plural positions.

[0015] Further, as mentioned later, the double coating structure is also employed preferably in a case where the connector portion supporting the external portion connecting terminal to the internal portion is integrally formed with the outer shell forming portion and the case portion. A severe dimensional accuracy is required in the connector portion since the connector portion generally has a structure of fitting to the mating connection so as to form a sealed state. However, when employing the double coating structure as mentioned above, the connector portion can be formed at the high pressure together with the outer shell forming portion and the case portion, and the dimensional accuracy of the connector portion can be enhanced.
[0016] Further, in a case where the double coating structure as mentioned above is employed, the coil coating forming portion is preferably provided with a concavo-convex portion for positioning the coil coating forming portion within the outer shell forming portion. The coil covered with the coil coating forming portion can be precisely arranged within a metal mold forming the outer shell forming portion with the use of the concavo-convex portion when forming the outer shell forming portion. Thus, a relative positional precision between the outer shell forming portion and the coil (coil forming portion) to be formed can be enhanced.
[0017] The element “concavo-convex portion” mentioned above can be formed by a concave portion, for example, a hole (preferably a plurality of holes), however, may be formed by a convex portion such as a projection (preferably a plurality of projections).
[0018] Further, the present invention may be provided with a connector portion which enables an electrical connection with the external portion by supporting the external portion connecting terminal to an internal portion thereof, and the connector portion may be integrally formed with the outer shell forming portion and the case portion.
[0019] Further, according to an aspect of the present invention, the case portion has an opening which allows the control board to be installed in the internal portion, the electrically operated valve includes a lid body which occludes the opening, a connector portion including an external portion connecting terminal which enables the electrical connection to the external portion, and a terminal coating forming portion which covers an intermediate portion of the external portion connecting terminal and supports the external portion connecting terminal to the connector portion, and the connector portion and the lid body are integrally formed in such a manner as to cover and support the terminal coating forming portion.
[0020] Further, in the aspect mentioned above, the terminal coating forming portion is preferably provided with the concavo-convex portion for positioning the terminal coating forming portion within the connector portion. This is because the external portion connecting terminal covered with the terminal coating forming portion can be accurately arranged within the metal mold forming the connector portion with the use of the concavo-convex portion and the external portion connecting terminal can be installed with a good positional precision when forming the connector portion. The element “concavo-convex portion” can be formed by the concave portion (for example, hole) or the convex portion (for example, projection) in the same manner as mentioned above.
[0021] Further, in the present invention, one or more press-in type projection pressed in a fixing hole is preferably provided as the support projection. This is because the board can be more securely supported within the case portion.
[0022] For the same reason, in the present invention, the support projection may have a rod-shaped projection main body, and a protruding portion which extends in a length direction of the projection main body and protrudes outward from an outer peripheral surface of the projection main body to come into contact with an inner peripheral surface of the fixing hole when the support projection is inserted into the fixing hole. In this case, at least a part of the protruding portion may be adapted to be crushed within the fixing hole when the support projection is inserted into the fixing hole, so that the support projection is pressed in the fixing hole.
[0023] Further, the support projection is preferably provided with two or more protruding portions, and the two or more protruding portions are preferably arranged radially with respect to a central

axis line of the projection main body. This is because the support projection can be easily inserted into the fixing hole when attaching the board and the board can be stably supported by the support projection.
[0024] Further, in the present invention, from the both point of view of workability when attaching the board, and certainty for support within the case portion (improvement of strength for supporting the board to the case portion), it is particularly preferably to employ the following aspect.
[0025] The case portion has an opening which allows the board to be installed in the internal portion, the electrically operated valve is provided with a lid body which occludes the opening, the support projection is provided with a step portion which comes into contact with a back surface of the board when slotting the board in the case portion while inserting the support projection into the fixing hole so as to stop the progress of the board toward a direction of a bottom surface of the case portion, the lid body is provided in an inner surface thereof with a pressing projection which comes into contact with a surface of the board when occluding the opening, and the board is held between the step portion and the pressing projection.
[0026] Further, in the aspect mentioned above, the pressing projection is preferably disposed so as to face to the support projection (in particular, the step portion), and the pressing projection is preferably adapted to have a tubular shape which can store a leading end portion of the support projection protruding out of the board. This is because the board can be firmly held. More specifically, on the assumption that a direction orthogonal to the board (length direction of the pressing projection and the support projection) is a “vertical direction”, and a direction parallel to the board is a “horizontal direction”, the pressing projection can be disposed at a position which does not face to the support projection and is deviated in the horizontal direction. In the structure as mentioned above, the board can be held. According to the aspect mentioned above, the pressing projection is adapted to come into contact with the surface of the board around the fixing hole (portion facing to the step portion and near the fixing hole), and the board can be more firmly held and supported by the pressing projection and the step portion which face to each other (overlap with each other looking from the vertical direction).
[0027] In the present invention, the board is supported within the case portion by the fixing hole which is provided in the board, and the support projection which is provided in the case portion. Plural sets of (for example, four sets of) the combination of the fixing hole and the support projection are preferably provided for securely supporting the board, that is, two or more fixing holes are preferably provided in the board, and two or more support projections are preferably provided in the case portion so as to correspond to the plurality of fixing holes. In this case, these two or more support projections are preferably adapted to include both the press-in type projection which is pressed in the fixing hole (that is, the support projection is inserted while being exposing to the pressure from the inner peripheral surface of the fixing hole when the support projection is put into from the support projection), and a loose-fit type projection which has a gap with respect to the inner peripheral surface of the fixing hole when being inserted into the fixing hole.
[0028] Because the support projection may be broken, for example, the support projection is snapped off due to a force applied when the board is attached or a vibration applied from a vehicle when the electrically operated valve is mounted to the vehicle and aging used, if all the support projections are formed by the press-in type projection. Further, if all the support projections are adapted to be pressed in the corresponding fixing holes, a high dimensional accuracy is demanded for a relative positional relationship between all the fixing holes and all the corresponding support projections, and a manufacturing cost is increased. Thus, the aspect as mentioned above is preferably employed.
[0029] Further, in the present invention, as a more specific structure of the case portion, the case portion may have an opening which allows the control board to be installed in the internal portion, the electrically operated valve may be provided with a lid body which occludes the opening, the lid body may be fixed to the case portion by melding both the case portion and the lid body (for

example, by welding by an infrared light) in a contact portion between the case portion and the lid body, and the opening may be occluded by the lid body.
[0030] Further, the case portion may be provided in an internal portion thereof with a terminal which performs an electrical connection to the control board, and a height of the terminal may be higher than a height of the support projection. According to the aspect mentioned above, it is possible to achieve a relative positioning between the terminal and the board (for example, a through hole disposed in the board) by inserting the support projection into the fixing hole of the board, and the terminal can be fitted to and inserted into the board (through hole) only by fitting and inserting the support projection to the fixing hole formed in the board and thereafter progressing the board toward the bottom surface of the case portion when attaching the board.
[0031] The electrical connection between the board (through hole) and the terminal is preferably achieved by a press-fit connection (terminal is adapted to be pressed in the through hole of the board as a press-fit pin) from the point of view of reducing the manufacturing step number. However, it is possible to employ the other methods of installing the board within the case portion and thereafter soldering.
[0032] According to the present invention, it is possible to attach the board to the electrically operated valve with a good workability, it is possible to reduce the parts number and the manufacturing step number of the electrically operated valve with board, and it is possible to reduce the manufacturing cost of the electrically operated valve.
[0033] The other objects, features and advantages of the present invention are clarified by the following description of embodiments according to the present invention described on the basis of the accompanying drawings. It is apparent for a person skilled in the art that the present invention is not limited to the following embodiments, but can be variously modified within the scope of claims. Further, same reference numerals in the drawings denote the same or corresponding portions.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0034] FIG. **1** is a vertical cross sectional view showing a valve close state of an electrically operated valve according to a first embodiment of the present invention.
[0035] FIG. **2** is a vertical cross sectional view showing a valve open state of the electrically operated valve according to the first embodiment.
[0036] FIG. **3** is a perspective view showing a manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which a coil cover covering a coil is formed (state after finishing a primary molding).
[0037] FIG. **4** is a perspective view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the coil forming the coil cover and an external portion connecting terminal forming a terminal cover are arranged within a metal mold (not shown) forming an outer shell cover.
[0038] FIG. **5** is a front elevational view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows the state in which the coil forming the coil cover and the external portion connecting terminal forming the terminal cover are arranged within the metal mold (not shown) forming the outer shell cover.
[0039] FIG. **6** is a vertical cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows the state in which the coil forming the coil cover and the external portion connecting terminal forming the terminal cover are arranged within the metal mold (not shown) forming the outer shell cover, looking from a lateral surface side.

[0040] FIG. **7** is a perspective view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state before the board is attached within the case.
[0041] FIG. **8** is a perspective view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the board is attached within the case.
[0042] FIG. **9** is a perspective view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the lid body is welded to the case.
[0043] FIG. **10** is a vertical cross sectional view showing a relationship among a fixing hole of the board, a press-in projection of the case, and a pressing projection of the lid body in an enlarged manner, for the electrically operated valve according to the first embodiment.
[0044] FIG. **11** is a front elevational view showing the relationship between the fixing hole of the board and the press-in projection of the case, for the electrically operated valve according to the first embodiment.
[0045] FIG. **12** is a vertical cross sectional view showing a relationship among the fixing hole of the board, a loose-fit projection of the case, and the pressing projection of the lid body in an enlarged manner, for the electrically operated valve according to the first embodiment.
[0046] FIG. **13** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state before the board is attached within the case.
[0047] FIG. **14** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which a support projection of the case begins to be inserted into the fixing hole of the board.
[0048] FIG. **15** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the support projection is inserted into the fixing hole of the board, and a terminal begins to be inserted into a through hole.
[0049] FIG. **16** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the terminal is inserted into the through hole, and the support projection begins to be pressed in the fixing hole.
[0050] FIG. **17** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the pressing operation of the support projection in the fixing hole is finished.
[0051] FIG. **18** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state before attaching a lid body.
[0052] FIG. **19** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the lid body is brought into contact with a front edge portion of the case.
[0053] FIG. **20** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state in which the attachment of the lid body is finished by welding the lid body to the case.
[0054] FIG. **21** is a cross sectional view showing the manufacturing step of the electrically operated valve according to the first embodiment, and shows a state before attaching the lid body to the case.
[0055] FIG. **22** is a vertical cross sectional view showing a valve close state of an electrically operated valve according to a second embodiment of the present invention.
[0056] FIG. **23** is a vertical cross sectional view showing a valve open state of the electrically operated valve according to the second embodiment.
[0057] FIG. **24** is a perspective view showing a manufacturing step of the electrically operated valve according to the second embodiment, and shows a state before attaching a board within a case.

[0058] FIG. **25** is a perspective view showing the manufacturing step of the electrically operated valve according to the second embodiment, and shows a state in which the board is attached within the case.

[0059] FIG. **26** is a perspective view showing the manufacturing step of the electrically operated valve according to the second embodiment, and shows a state in which a lid body is attached to the case.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0060] A description will be given of an electrically operated valve according to a first embodiment of the present invention with reference to FIGS. **1** to **21**. Each of the drawings appropriately shows two-dimensional coordinates or three-dimensional coordinates which represent a longitudinal direction, a horizontal direction and a vertical direction and are orthogonal to each other, and the following description will be given on the basis of these directions (same applies to a second embodiment mentioned later).

[0061] An electrically operated valve **11** according to a first embodiment of the present invention is an electrically operated valve which is preferably used for regulating a flow rate of a refrigerant in a refrigeration cycle apparatus, for example, an air conditioning machine, and is provided with a valve main body **12** having a valve chamber **13** in an internal portion thereof and also having an inflow passage **16** which allows the refrigerant to flow into the valve chamber **13** and an outflow passage **15** which allows the refrigerant to flow out of the valve chamber **13**, a valve seat **14** which is formed in an opening portion for the valve chamber **13** in the inflow passage **16**, a valve body **17** which changes a passing quantity (flow rate) of the refrigerant by moving forward and backward (moving upward and downward) with respect to the valve seat **14** between a valve close state (refer to FIG. **1**) coming into contact with the valve seat **14** and a valve open state (refer to FIG. **2**) being away from the valve seat **14**, an electric motor **23** which drives the valve body **17**, a coupling member **19** which couples the electric motor **23** to the valve main body **12**, a printed board (control board) **46** which mounts a controller (not shown) controlling the electric motor **23** thereon, a case (board storage portion/case portion) **43** which stores the board **46**, a lid body **47** which occludes a front opening of the case **43**, a connector **44** which has an external portion connecting terminal **45** electrically connecting to the external portion, and a can (sealed container) **22** which covers an upper opening **12***a* of the valve main body **12** communicating with the valve chamber **13** and forms a sealed space together with the coupling member **19**. In the present embodiment, the refrigerant is flowed into from the inflow passage **16** and the refrigerant is flowed out of the outflow passage **15**. However, it goes without saying that the electrically operated valve **11** can be used in a case where a flowing direction of the refrigerant is an inverse direction.

[0062] The electric motor **23** is constructed by a stepping motor including a stator **25** which is arranged in an outer side of the can **22**, and a magnet rotor (hereinafter, refer simply to “rotor”) **24** which is arranged in an inner side of the can so as to be rotatable and slidable in a vertical direction. Further, the stator **25** includes a yoke **26**, a bobbin **27** and a coil **28**.

[0063] The stator **25** and the can **22** are covered by an outer shell cover (outer shell forming portion) **42** made of a synthetic resin. Further, the outer shell cover **42** is provided in an inner side thereof with a coil cover (coil coating forming portion) **41** which covers the coil **28** and is made of a synthetic resin, and a terminal cover (terminal coating forming portion) **63** which covers and supports an intermediate portion of the external portion connecting terminal **45**.

[0064] The external portion connecting terminal **45** has a leading end portion (end portion in the connector side) **45***a* which protrudes toward an internal space **44***a* of the connector **44** and allows an electrical connection to the external portion, and a base end portion (end portion in the board side) **45***b* which is electrically connected to the board **46**. An intermediate portion corresponding to a portion between the leading end portion **45***a* and the base end portion **45***b* is covered by the terminal cover **63**.

[0065] Further, each of the covers (coil cover **41**, terminal cover **63** and outer shell cover **42**) is formed according to an injection molding. In the order of formation, first of all, the coil cover **41** and the terminal cover **63** (fine dot pattern is applied to a cross section of each of the covers **41** and **63** in FIGS. **1** and **2**) are independently formed (this step is called as “primary molding”), and the outer shell cover **42** (rough dot pattern is applied to a cross section in FIGS. **1** and **2**) is formed in such a manner as to cover each of the covers (coil cover **41** and terminal cover **63**) (this step is called as “secondary molding”).

[0066] Here, in the primary molding, a positioning concave portion used when performing the secondary molding is formed in each of the coil cover and the terminal cover.

[0067] More specifically, FIG. **3** shows a state in which the coil after the primary molding is reversed upside down so that a lower surface side of the coil cover **41** is visible. However, as shown in FIG. **3**, in the primary molding, a plurality of (four in the present embodiment) notched concave portions **64** are formed as concave portions for positioning in an inner peripheral edge portion of the bottom surface of the coil cover **41**. Further, FIGS. **4** to **6** shows a state in which the coil (which is covered with the coil cover **41**) after the primary molding and the external portion connecting terminal **45** (which is covered with the terminal cover **63** in an intermediate portion) are arranged within a metal mold **66** forming the outer shell cover **42**. A plurality of (three in the present embodiment) holes **65** are formed as concave portions for positioning in the terminal cover **63**. The metal mold **66** forming the outer shell cover **42** is provided in an internal portion thereof with engagement portions (for example, engagement projection (not shown)) which engage respectively with the notched concave portion and the hole.

[0068] Therefore, according to the present embodiment, it is possible to accurately arrange the coil **28** covered with the coil cover **41** and the external portion connecting terminal **45** provided with the terminal cover **63** within the metal mold **66** forming the outer shell cover **42** by the use of the concave portions **64** and **65** for positioning. Accordingly, it is possible to install the coil **28** and the external portion connecting terminal **45** within the outer shell cover **42** while accurately keeping a relative positional relationship between the coil **28** and the external portion connecting terminal **45**. Further, in the present embodiment, the plurality of concave portions **64** and **65** for positioning are disposed respectively in the coil cover **41** and the terminal cover **63**. Thus, it is possible to accurately position the coil **28** and the external portion connecting terminal **45** within the metal mold **66** performing the secondary molding, and it is possible to securely prevent a positional displacement from being generated in the coil **28** and the external portion connecting terminal **45** by the resin flowing into the metal mold **66** when performing the secondary molding.

[0069] When performing the secondary molding, the box-shaped case **43** storing the board **46** and the connector **44** are integrally formed with the outer shell cover **42**. The resin materials constructing the coil cover **41**, the terminal cover **63**, the outer shell cover **42** (including the case **43** and the connector **44**) and the lid body **47** mentioned later can be set to different kinds of materials, however, preferably employ the same material for enhancing a bondability between the resins.

[0070] In the present embodiment, the case **43** is disposed in a lateral side (front side) of the stator **25**, and the connector **44** is disposed in an upper surface portion of the case **43**. The board **46** is stored in the internal portion **43***a* of the case **43** through the front opening of the case **43**, and the coil **28** is electrically connected to the external portion connecting terminal **45** via the board **46**. This is because of enabling power feeding from an external power supply (not shown) to the coil **28**. Further, a controller including a pulse generator and a motor driving circuit is mounted to the board **46**. Further, in a case where a magnetic sensor detecting rotation of the rotor **24** is provided in the electrically operated valve **11**, a computing apparatus computing an angle of rotation of the rotor **24** and an opening degree of the valve on the basis of an output signal from the magnetic sensor may be mounted to the board **46**.

[0071] Each of the case **43** and the connector **44** is constructed as a sealable structure. More specifically, in the connector **44**, the internal portion **44***a* of the connector **44** comes to a sealed state by fitting a mating connector to a front end portion thereof. The mating connector means a connector having a specific shape which can be fitted and connected to a connector of the electrically operated valve. More specifically, the connector is generally manufactured in such a manner as to have a specific shape (fixed shape which is different for every user and is previously determined) adapted to (conform to) the user side of the electrically operated valve, and the connector comes to the sealed state by fitting the mating connector having the specific shape as mentioned above thereto. The sealed state means a state in which any moisture does not infiltrate into the internal portion **44***a* of the connector **44** in a normal use state of the electrically operated valve **11**. In the present embodiment, the internal portion **44***a* of the connector **44** is cut off from the external space (atmosphere side).

[0072] In the meantime, the case **43** comes to a sealed state by welding the resin lid body **47** to the front opening and closing the front opening. The attachment of the lid body to the case will be later described in detail together with a structure for supporting the board.

[0073] Further, the electrically operated valve **11** according to the present embodiment is provided with a rod-shaped valve stem **18** which extends in a vertical direction from an internal portion of the rotor **24** toward the valve chamber **13** along a central axis line A. The valve stem **18** has a columnar barrel portion **18***a*, and an upper small-diameter portion **18***b* which is coaxially formed in an upper end portion of the barrel portion **18***a* in succession to the barrel portion **18***a* and has a small outer diameter. Further, the valve body **17** is integrally provided in a lower end of the valve stem **18** (barrel portion **18***a*). The rotor **24** is arranged in an inner side of the can **22** so as to be rotatable and slidable in the vertical direction, and the valve is opened and closed by the integral movement in the vertical direction of the valve step **18** provided with the valve body **17** in the lower end and the rotor **24**.

[0074] A valve stem holder **31** is disposed in an inner side of the rotor **24**. The valve stem holder **31** has a cylindrical shape which is closed in an upper end, and a support ring **33** is fixed to an upper end portion of the valve stem holder **31** according to a caulking. Further, the rotor **24** and the valve stem holder **31** are integrally connected via the support ring **33**. A female screw portion **31***a* is formed on an inner peripheral surface of the valve stem holder **31**. The female screw portion **31***a* constructs a transfer mechanism (screw feeding mechanism) which is threadably mounted on a male screw portion **36***a* of a guide bush **36** mentioned later and converts the rotation of the electric motor **23** into a linear motion to transfer to the valve stem **18**.

[0075] The upper small-diameter portion **18***b* of the valve stem **18** passes through the valve stem holder **31**, and a push nut **34** for retaining is attached to an upper end portion of the upper small-diameter portion **18***b*. The valve stem **18** is biased downward by a compression coil spring **35** which is disposed between the valve stem holder **31**, and a step portion between the barrel portion **18***a* and the upper small-diameter portion **18***b* in the valve stem **18**. Therefore, the valve stem **18** is regulated its relative movement in the vertical direction to the valve stem holder **31** by the push nut **34** and the compression coil spring **35**, and moves upward and downward together with the valve stem holder **18**.

[0076] The coupling member **19** is a tubular member having a large-diameter hole **19***a* and a small-diameter hole **19***b* which are through holes communicating with each other. The large-diameter hole **19***a* passes through an upper center portion of the coupling member **19**, and has a large diameter so that the guide bush **36** mentioned later can be fitted and inserted from the above. The small-diameter hole **19***b* passes through a lower center portion of the coupling member **19** and has a small diameter. Further, the cylindrical can **22** which is covered and not bottomed (in which a bottom surface is opened and a top surface is occluded) is joined to an outer peripheral surface of the upper end portion of the coupling member **19** via a ring-shaped base plate **21**.

[0077] The guide bush **36** is fixed to the large-diameter hole **19***a* in the upper portion of the coupling member **19**. The guide bush **36** has a large-diameter cylindrical portion **36***a* which has a large outer diameter, and a small-diameter cylindrical portion **36***b* which is coaxially formed in an

upper end portion of the large-diameter cylindrical portion **36***a* in succession to the large-diameter cylindrical portion **36***a* and has a small outer diameter. A male screw portion **36***c* threadably mounted on the female screw portion **31***a* of the valve stem holder **31** is formed on an outer peripheral surface of the small-diameter cylindrical portion **36***b*. The guide bush **36** is connected to the coupling member **19** by pressing the large-diameter cylindrical portion **36***a* in an inner side of the coupling member **19**. Further, the barrel portion **18***a* of the valve stem **18** passes through the small-diameter hole **19***b* of the coupling member **19**.
[0078] Further, an upper stopper body **32** is disposed in the valve stem holder **31**, and a lower stopper body **37** is disposed in the large-diameter cylindrical portion **36***a* of the guide bush **36**. The stopper bodies **32** and **37** are provided for determining a lower limit position of the valve stem holder **31**, and when the valve stem holder **31** moves down by rotating to reach the lower limit position, the upper stopper body **32** comes into contact with the lower stopper body **37** and further rotation of the valve stem holder **31** is regulated.
[0079] The mechanism for transferring the drive force of the electric motor **23** to the valve body **17** can employ various mechanisms in addition to the above mechanism, and is not limited to the above structure.
[0080] A description will be given below of a motion of the electrically operated valve **11** according to the present embodiment.
[0081] When an electric current is supplied to the stator **25** (coil **28**) in such a manner that the rotor **24** turns in one direction from the valve close state shown in FIG. **1**, the valve stem holder **31** connected to the rotor **24** turns together with the rotor **24**. Since the female screw portion **31***a* threadably mounted on the male screw portion **36***c* formed on the outer peripheral surface of the small-diameter cylindrical portion **36***b* of the guide bush **36** is formed on the inner peripheral surface of the valve stem holder **31**, the rotation of the rotor **24** (valve stem holder **31**) is converted into the liner motion in the vertical direction on the basis of an interaction of the male screw portion **36***c* and the female screw portion **31***a*, and the valve stem holder **31** moves upward. As a result, the rotor **24** connected to the valve stem holder **31**, and the valve stem **18** regulating its relative movement to the valve stem holder **31** also move upward together with the valve stem holder **31**. The valve body **17** disposed in the lower end of the valve stem **18** is away from the valve seat **14** accompanied with the upward movement of the valve stem **18**, so that the refrigerant flowing into from the inflow passage **16** is going to flow out of the outflow passage **15** through the valve chamber **13** (refer to FIG. **2**). A passing quantity (refrigerant flow rate) of the refrigerant can be adjusted on the basis of an amount of rotation of the rotor **24**.
[0082] On the contrary, when the electric current is supplied to the stator **25** (coil **28**) in such a manner that the rotor **24** turns in a direction opposite to the one direction from the valve open state, the rotation of the rotor **24** (valve stem holder **31**) is converted into the linear motion in the vertical direction on the basis of the interaction of the female screw portion **31***a* and the male screw portion **36***c*. As a result, the valve stem holder **31** moves downward together with the rotor **24** and the valve stem **18**. Thus, the valve body **17** moves downward toward the valve seat **14**, and a flow passage between the inflow passage **16** and the outflow passage **15** is cut off when the valve body **17** comes into contact with the valve seat **14**, thereby coming to the valve close state (refer to FIG. **1**).
[0083] A description will be given of the structure for supporting the board **46** to the case **43**, and the attachment of the lid **47** to the case **43** with reference to FIGS. **7** to **21**.
[0084] The present embodiment is provided with support projections **51** and **58** which are integrally formed with the case **43**, a fixing hole **67** which is pierced in the board **46**, and a pressing projection **61** which is formed in a back surface (inner surface) of the lid body **47** occluding the front opening of the case **43**, as means for supporting the board **46** in the internal portion of the case **43**, and controls the movement of the board **46** in a vertical direction by holding the board **46** by means of a step portion **57** and a pressing projection **61** disposed in the support projections **51** and **58** as well as controlling the movement of the board **46** in a horizontal direction by the support

projections **51** and **58** fitted and inserted to the fixing hole **67**, thereby supporting the board **46** to the internal portion of the case **43**.

[0085] Describing each of the means in particular, the internal portion of the case **43** is provided with a plurality of (four in the present embodiment) support projections **51** and **58** which rise vertically (forward) from the bottom surface portion of the case **43**. Further, the board **46** is provided with a plurality of (four in the present embodiment so as to correspond to the support projections) fixing holes **67** to which the support projections **51** and **58** are fitted and inserted. Further, a back surface (inner surface) of the lid body **47** occluding the front opening of the case **43** is provided with the pressing projection **61** which comes into contact with the surface of the board **46** when the board **46** is installed within the case **43** and the lid body **47** is joined to the case **43**.

[0086] The support projections **51** and **58** are constructed by two press-in projections (press-in type projections) **51** and two loose-fit projections (loose-fit type projections) **58**. These support projections **51** and **58** are integrally formed with the case **43** according to the secondary molding which forms the outer shell cover **42**. Further, the step portions **57** expanding horizontally toward an outer side from a peripheral edge of the fixing hole **67** are formed in the bottom surface portions of the support projections **51** and **58**. These step portions **57** is adapted to perform a function for stopping the progress of the board **46** (receiving the back surface of the board **46**) by the contact of the back surface of the board **46** when inserting the support projections **51** and **58** into the fixing holes **67** and progressing the board **46** to the direction of the bottom surface of the case **43** and holding the board **46** in cooperation with the pressing projections **61** formed in the back surface of the lid body **47**.

[0087] As shown in FIGS. **10** and **11** in an enlarged manner, the press-in projection **51** has a projection main body **51***a* which has an approximately columnar shape, and a protruding portion **55** which protrudes outward from an outer peripheral surface of the projection main body **51***a*. The protruding portion **55** extends in a vertical direction (longitudinal direction) which corresponds to a length direction of the press-in projection **51**, and has a triangular transverse cross sectional shape (refer to FIG. **11**) which is sharpened toward an outer direction from an outer peripheral surface of the projection main body **51***a*.

[0088] A ridge line portion (apex portion of the triangle when viewed from the transverse cross section) of the protruding portion **55** protrudes outward from an inner peripheral surface of the fixing hole **67** in the board **46**. When putting a portion where the protruding portion **55** is formed (this portion is called as "press-in portion") into the fixing hole **67** in the longitudinal direction of the press-in projection **51**, the ridge line portion (hatched portion in FIG. **11**) **53** of the protruding portion **55** is crushed within the fixing hole **67**. As a result, the press-in projection **51** is fixed within the fixing hole **67**.

[0089] Further, a plurality of (three in the present embodiment) protruding portions **55** are formed in each of the press-in projections **51**, and these three protruding portions **55** are arranged in such a manner that the protruding portions **55** are radially arranged with respect to a central axis line B of the press-in projection **51** (angles formed by the adjacent protruding portions **55** are equal, that is, 120 degrees in the present embodiment). This is because of preventing the board **46** from being exposed to the force in the horizontal direction from the protruding portion **55** when pressing the press-in projection **51** in the fixing hole **67** of the board **46** to cause a lateral displacement and stably supporting the board **46** within the case **43**. The number of the protruding portions **55** can be set to be, for example, four, or two or less or five or more.

[0090] Further, a leading end (front end) portion of the projection main body **51***a* is formed as a small-diameter portion **52** in which an outer diameter is made small in such a manner as to be easily inserted into the fixing hole **67** of the board **46**, and a taper surface **54** is formed from the small-diameter portion **52** to the press-in portion **53**, the taper surface having an outer diameter which is increased little by little. This is because the board **46** (fixing hole **67**) is smoothly guided to the press-in portion **53**. Further, with regard to the protruding portion **55**, a front end portion (leading end portion in the length direction) thereof is formed into a taper shape in such a manner that a height of the ridge line portion **56** is increased little by little toward a rear side in correspondence to the taper surface **54**.

[0091] In the meantime, the loose-fit projection **58** is provided in a bottom surface portion with a step portion **57** which performs the same function as the step portion **57** of the press-in projection **51** as shown in FIG. **12**, and has a columnar projection main body **58***a* which rises vertically (forward) from the step portion **57**. The projection main body **58***a* has such an outer diameter that a fixed gap S is formed with respect to an inner peripheral surface of the fixing hole **67** in the board **46**.

[0092] The support projections **51** and **58** are arranged in four corners of the case **43** having a rectangular front shape. In more detail, the press-in projections **51** are arranged in both end portions of one diagonal line of the case **43**, and the loose-fit projections **58** are arranged I both end portions of the other diagonal line of the case **43**, respectively. All the support projections **51** and **58** can be constructed by the press-in projection **51**. However, if all the support projections are constructed by the press-in projection **51**, a high dimensional accuracy is demanded for the relative positional relationship between the support projection **51** and the fixing hole **67** as mentioned above, and the manufacturing cost is increased. Further, due to the force application when attaching to the board or the vibration applied when being mounted to the vehicle and aging used, there is a risk that the support portion of the board **46** is damaged, for example, snapping off of the support projection **51**. Therefore, the loose-fit projection **58** is included in the present embodiment.

[0093] The pressing projection **61** in the back surface of the lid body **47** has a cylindrical shape which can store the leading end portions of the support projections **51** and **58**. A height (dimension in a length direction) of the pressing projection **61** is set to a height that a fixed gap is formed from the surface of the board **46** (refer to FIG. **19** mentioned later) when the lid body **47** is put on the front opening of the case **43** and the peripheral edge portion of the lid body **47** is brought into contact with the peripheral edge portion of the front opening, and the pressing projection **61** comes into contact with the surface of the board **46** (refer to FIG. **20** mentioned later) when the lid body **47** is welded (melt joined) to the case **43**. The reason why the pressing projection **61** is constructed as the above height structure will be later mentioned in detail.

[0094] A description will be given of the installation of the board **46** to the case **43** and the attachment of the lid body **47** step by step according to a manufacturing step with reference to FIGS. **13** to **20**.

[0095] As shown in FIGS. **13** to **14**, when the board **46** is set horizontally (in parallel to the bottom surface of the case **43**) and is slotted in the case **43** from the front opening, the leading end portions of the support projections (press-in projection **51** and the loose-fit projection **58**) in the case **43** are first put into the fixing hole **67** of the board **46** (refer to FIG. **15**).

[0096] Next, when the board **46** is going to be progressed toward the bottom surface of the case **43** from this state, the terminals **45***b* and **62** disposed within the case **43** are going to be put into a through hole **68** disposed in the board **46**. Since the leading end portions of the support projections **51** and **58** are put into the fixing holes **67** in the step shown in FIG. **15** at this time, the relative positioning between the board **46** and the case **43** in the horizontal direction is approximately finished (the board **46** is not positionally displaced in the horizontal direction beyond a smaller gap in the gap between the small-diameter portion **52** in the leading end of the press-in projection and the inner peripheral surface of the fixing hole **67**, and the gap S between the loose-fit projection **58** and the fixing hole **67**), and it is possible to smoothly insert the terminals **45***b* and **62** into the through hole **68** (refer to FIG. **16**). Further, the pressing-in operation of the press-in projection **51** (insertion of the press-in portion **53** into the fixing hole **67**) is simultaneously started.

[0097] Further, when the board **46** is advanced rearward (toward the bottom surface in the case) until the back surface of the board **46** runs into the step portion **57** as shown in FIG. **17**, the arrangement of the board **46** is finished.

[0098] The electrical connection between the terminals **45***b* and **62** and the board **46** (through hole **68**) may be achieved by a press-fit connection in which the terminals **45***b* and **62** are formed as the press-fit pin (electrical connection is achieved by pressing in the through hole **68**), or may be achieved by soldering after arranging the board **46**.
[0099] After the board **46** is arranged within the case **43**, the lid body **47** is attached to the case **43**. As shown in FIGS. **18** to **19**, the lid body **47** is put on the case **43** in such a manner as to close the front opening, and the peripheral edge portion of the back surface in the lid body is brought into contact with the front edge portion of the case **43**. In this state, the lid body **47** is pressed against the bottom surface of the case **43** while irradiating an infrared light onto a contact surface of both the elements (front edge portion of the case and the peripheral edge portion of the lid body) and welding the contact surface between the case **43** and the lid body **47**. The lid body **47** progresses toward the bottom surface of the case **43** accompanied with the welding of the contact surface. However, when the pressing projection **61** in the back surface of the lid body runs into the surface (front surface) of the board **46** as shown in FIG. **20**, the lid body **47** does not progress any more. Further, when the welding is finished in this state, the board **46** is firmly supported within the case **43** in a state in which the board is held by the step portion **57** and the pressing projection **61**. The weld surface of the lid body **47** and the case **43** is denoted by reference numeral **69** in FIG. **20**.
[0100] As described above, according to the present embodiment, it is possible to install the board **46** into the case **43** and attach the lid body **47** only by a comparatively simple operation of pressing the board **46** toward the bottom surface of the case **43** and welding the lid body **47**. Further, the welding depth of the case **43** and the lid body **47** (how far the case **43** and the lid body **47** is welded in the vertical direction (longitudinal direction)) can be set by the height (length in the longitudinal direction) of the pressing projection. Therefore, according to the present embodiment, it is possible to prevent the airtightness of the case **43** from being insufficient due to insufficient welding depth or prevent an excessive welding from being generated, by setting the height of the pressing projection **61** appropriately. As a result, it is possible to keep a welding quality of the case **43** fixed or good when mass producing the electrically operated valve **11**. Further, the joining work of the lid body **47** can be evenly finished by the contact of the pressing projection **61** with the surface of the board **46**. Therefore, it is possible to efficiently perform the attaching work of the lid body **47**.
[0101] Further, FIG. **21** shows a structure which prevents the board **46** from being erroneously attached. As shown in FIG. **21**, in the present embodiment, notch portions **81** and **82** having different shapes are respectively formed in a left edge portion and a right edge portion of the board **46**, and a projection **83** fitted to one notch portion (wide notch portion in the left side in the present embodiment) **81** when arranging the board **46** correctly is formed in the internal portion of the case **43**. In a case where the board **46** is arranged inside out, the narrower notch portion **82** in the right side of FIG. **21** comes to the left side, and the projection **83** can not be fitted to the notch portion **82**. As a result, it is possible to immediately know that the board **46** is inside out. Thus, it is possible to prevent the board **46** from being erroneously arranged inside out when attaching the board.
Second Embodiment
[0102] A description will be given of an electrically operated valve according to a second embodiment of the present invention with reference to FIGS. **22** to **26**. In the following description, same reference numerals are attached to the same structures as those of the electrically operated valve **11** according to the first embodiment, a redundant description will be omitted, and the description will be given mainly of different points.
[0103] As shown in FIGS. **22** to **26**, an electrically operated valve **71** according to the present embodiment is adapted to move the valve body **17** upward and downward by the electric motor (stepping motor) **23** to regulate the flow rate of the refrigerant in the same manner as the electrically operated valve **11** according to the first embodiment, and is provided with the board **46** which controls the electric motor **23**. However, the board **46** is arranged in the upper surface

portion of the electric motor **23** (stator **25**) so as to horizontally extend, and the case **43** storing the board **46** is formed in the upper surface portion of the electric motor **23**.

[0104] The present embodiment is provided with the outer shell cover **42** (to which a rough dot pattern is applied to a cross section in FIGS. **7** and **8**) which covers the stator **25** of the electric motor **23** and the can **22**, and the coil cover **41** (to which a fine dot pattern is applied to a cross section in FIGS. **7** and **8**) which is arranged in the inner side of the outer shell cover **42** and covers the coil **28** in the same manner as the electrically operated valve **11** according to the first embodiment. However, as is different from the electrically operated valve **11** according to the first embodiment, the connector **44** is not integrally formed with the outer shell cover **42**, but is formed as a lid body with connect or (lid body forming portion with connector portion) **72** in a different step and is integrally formed with the lid body **47** which occludes the upper opening of the case **43**.

[0105] The lid body **72** with connector is manufactured via two stages of forming steps. First of all, a first step (primary molding) forms the terminal cover (terminal coating forming portion) **63** which covers and supports the intermediate portion of the external portion connecting terminal **45** and is made of resin, and a successive second step (secondary molding) integrally forms the connector **44** and the lid body **47** as the lid body **72** with connector in such a manner that the terminal cover **63** is covered, and the terminal cover **63** is supported between the connector **44** and the lid body **47**, in other words, such that one end portion of the terminal cover **63** protrudes to the connector internal portion **44***a*, and the other end portion of the terminal cover **63** protrudes downward from the lower surface of the lid body **47**.

[0106] The lid body **72** with connector is welded to the upper edge portion of the case **43** in such a manner as to occlude the upper opening of the case **43** after installing the board **46** in the case internal portion **43***a*. The electrical connection between the external portion connecting terminal **45** and the board **46** may be achieved by a press fit connection. More specifically, with the use of the base end portion (end portion in the board side) **45***a* of the eternal portion connecting terminal **45** as a press fit pin, the press-fit pin is adapted to be pressed in the through hole **68** of the board **46** when putting the lid body **72** with connector on the upper surface of the case **43**.

[0107] The structure for supporting the board **46** to the case **43** is the same as the first embodiment. More specifically, the press-in projection **51** and the loose-fit projection **58** are provided as the support projection in the case **43** side, and the step portion **57** supporting the board **46** from the lower surface side is formed in the bottom surface portion of each of the support projections **51** and **58**. The board **46** has the fixing hole **67** to which the support projections **51** and **58** are fitted and inserted. In the meantime, the pressing projection **61** is disposed in the back surface (lower surface) in the lid body **72** with connector side.

[0108] When installing the board **46** within the case **43** while fitting and inserting the support projections **51** and **58** to the fixing hole **67** (refer to FIGS. **24** to **25**), and thereafter welding the lid body **72** with connect or to the upper surface of the case **43** (refer to FIG. **26**), the board **46** is controlled by the press-in projection **51** in the movement in the horizontal direction, and is controlled in the movement to the vertical direction by being held by the step portion **57** and the pressing projection **61**, thereby being fixed within the case **43**.

Claims

1. An electrically operated valve comprising: a valve main body having a valve chamber communicated with an inflow passage introducing a refrigerant and an outflow passage discharging the refrigerant; a valve body for changing a flow rate of the refrigerant by moving forward and backward with respect to a valve seat within the valve chamber between a valve close state seating on the valve seat and a valve open state separating from the valve seat; an electric motor for driving the valve body; a control board on which an electronic component controlling the electric motor is mounted; a board storage portion which includes a case portion storing the control board; and a stator of the electric motor; wherein the control board has a fixing hole to fix the control board to an inner side of the case portion, wherein the case portion has in an internal portion thereof a support projection supporting the control board within the case portion by being inserted into the fixing hole, wherein the support projection is integrally formed with the case portion, wherein the case portion is integrally formed with the stator, wherein the case portion has an opening which allows the control board to be installed in the internal portion, wherein the electrically operated valve comprises: a connector portion including an external portion connecting terminal which enable the electrical connection to the external portion; and a terminal coating forming portion which covers an intermediate portion of the external portion connecting terminal and supports the external portion connecting terminal to the connector portion, and wherein the terminal coating forming portion is provided with a concavo-convex portion for positioning the terminal coating forming portion within the connector portion.

2. An electrically operated valve comprising: a valve main body having a valve chamber communicated with an inflow passage introducing a refrigerant and an outflow passage discharging the refrigerant; a valve body for changing a flow rate of the refrigerant by moving forward and backward with respect to a valve seat formed within the valve chamber between a valve close state seating on the valve seat and a valve open state separating from the valve seat; an electric motor which drives the valve body; and a stator of the electric motor; wherein the case portion is integrally formed with the stator, wherein a coil coating forming portion covering a coil provided within the stator is disposed in an inner side of the stator, and wherein the coil coating forming portion is provided with a concavo-convex portion for positioning the coil coating forming portion within the stator.